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ORIGINAL ARTICLE

LC–MS metabolomics of urine reveals distinct profiles for non-muscle-invasive and muscle-invasive bladder cancer

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Abstract

Purpose Bladder cancer (BC) is among the most frequent malignancies worldwide. Novel non-invasive markers are needed to diagnose and stage BC with more accuracy than invasive procedures like cystoscopy. To date, no study has identified urine metabolites characteristic of all BC stages. To discover novel urine metabolomic profiles to diagnose and stage non-muscle-invasive (NMIBC) and muscle-invasive (MIBC) patients using mass spectrometry-based metabolomics.

Methods We prospectively recruited 198 BC patients and 98 age- and sex-matched healthy volunteers without evidence of renal or bladder condition confirmed by ultrasound, from whom we collected a first morning urine sample (before surgery in patients). In a discovery stage, an untargeted metabolomic analysis was conducted in urine samples of a selection of 64 BC patients (19 TaG1, 11 TaG3, 20 T1G3, 12 T2G3, 1 T2G2, 1 T3G3) and 20 controls to identify dysregulated metabolites. Next, after exhaustive multivariate analysis, confirmed dysregulated metabolites were validated in an independent cohort of 134 BC patients (19 TaG1, 62 TaG2, 9 TaG3, 15 T1G2, 16 T1G3, 4 T2G2, 9 T2G3) and 78 controls.

Results We validated *p*-cresol glucuronide as potential diagnostic biomarker for BC patients compared to controls (AUC = 0.79). For NMIBC, *p*-cresol glucuronide was valuable as staging biomarker (AUC = 0.803). And among NMIBCs, *p*-coumaric acid may be a potential specific staging biomarker for the TaG1 NMIBC; however, future validation experiments should be conducted once the precise version of the standard is commercially available. Remarkably, for MIBC we validated spermine as potential specific staging biomarker (AUC = 0.882).

Conclusion Ours is the first metabolomics study conducted in urine of a thoroughly characterized cohort comprising all stages of NMIBC, MIBC and healthy controls in which we identified non-invasive diagnostic and staging biomarkers. These may improve BC management, thus reducing the use of current harmful diagnostic techniques.

Keywords Bladder cancer · Biomarkers · Metabolomics · Non-invasive · *p*-Coumaric acid · *p*-Cresol glucuronide · Spermine · Urine

Introduction

Bladder cancer (BC) is among the most frequent malignancies worldwide. BC is responsible for 3% of all malignant tumors in adults and is the most lethal urological malignancy. Indeed, according to the GLOBOCAN 2020 study, BC accounts for 573,278 new cases worldwide and 212,536 deaths annually [1, 2]. Moreover, by the end of the decade, the disease is expected to account for > 3% of all cancer-related medical expenses. In fact, BC is the

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tumor with the highest monitoring costs and the most expensive cancer to treat, with the cost of the muscle-invasive subtype approaching $150,000 per capita [3, 4].

At present, bladder ultrasound, cytology and cystoscopy are the gold standards for BC detection and follow-up. However, cystoscopy is an invasive examination with potential secondary effects. Cytology is a non-invasive test that can identify tumor cells shed in urine but presents low sensitivity, especially in low-grade vesical tumors [5]. Novel non-invasive markers are needed to early diagnose and stage BC patients what may prevent the frequent performance of this detrimental technique, may reduce the resulting high rate of false positives and negatives and may improve the therapeutic impact. To overcome these drawbacks, in the last years, much effort has been centered into the development of novel urinary biomarker tests for BC [6], some of which are FDA- and EMA-approved like the Bladder Tumor Antigen (BTA) stat and BTA TRAK (Polymedco Inc., Cortlandt, NY, USA), NMP22 kit and NMP22 BladderChek (Alere, Waltham, MA, USA), ImmunoCyt/uCyt + assay (Scimedx Inc., Denville, NJ, USA), UroVysion (Vysis, Abott Molecular Inc., Chicago, IL, USA) [7]. However, none of these are recommended by international guidelines and used in daily clinical practice to replace cystoscopy since they fail to diagnose 18–43% of BCs and yield a 12–26% false positive rate [7]. Recently, novel urinary biomarker tests like Xpert bladder cancer (Cepheid; Sunnyvale, California, USA), Adxbladder (Arquer Diagnostics; Sunderland, UK), Bladder EpiCheck (Nucleix; Rehovot, Israel), Uromonitor (U-Monitor; Porto, Portugal), Cxbladder Monitor, and Triage and Detect (Pacific Edge; Dunedin, New Zealand) proved a high diagnostic accuracy supporting their utility for NMIBC surveillance, thus avoiding up to 740 cystoscopies in the 1000 patients studied [8].

Provided that urine is in direct contact with bladder epithelial cells, in the presence of a tumor BC cells may release specific metabolites to this bio-fluid. Previous studies in urine have compared the metabolomic profile of non-muscle-invasive (NMIBC) patients and controls [9–17] or have identified a metabolomic profile able to discriminate low- and high-grade BC patients [9, 10, 15, 17], but none studied individually each NMIBC stage or muscle-invasive (MIBC) patients. Additional studies focused on identifying recurrences in low-grade NMIBC [16] or were conducted in more invasive samples [18–21]. All in all, despite profuse studies in the recent years have addressed the aim to describe a urine metabolomic profile with diagnostic value for BC, to the best of our knowledge, none has described urine metabolomic profiles specific of each NMIBC stage and, remarkably, of MIBC stages. Accordingly, this became one of the main goals of the present study.

Materials and methods

Study subjects

One hundred and ninety eight BC patients and 98 age- and sex-matched healthy volunteers (control group) who underwent an ultrasound scan to rule out the presence of urological malignancies or other alterations were recruited between May 2016 and April 2018 at La Fe University and Polytechnic Hospital (Valencia, Spain). BC patients were recruited before surgery being elected independently of being a de novo diagnosis or a recurrent tumor. Controls were hospital staff members, unrelated friends or family of staff members or individuals who attended the urology service of our hospital for a consult totally unrelated to bladder, kidney or prostate cancer. These were periodically recruited to match patients. All were clinically followed-up until May 2020.

Pre-operative clinical staging was performed through physical examination, urine cytology and CT scans of the chest, abdomen, and pelvis (in case of invasive BC). The tumor histological classification was done according to grade in the WHO 1973 classification. Demographic and clinical data were collected.

The exclusion criteria were lack of informed consent, absence of histological confirmation and presence of other malignancies. Additional exclusion criteria for the controls were chronic or acute cystitis and upper urinary tract malignancies.

Informed consent was obtained from all participants according to protocols approved by the ethics review board at La Fe University and Polytechnic Hospital. The study was performed according to the declaration of Helsinki, as amended in Edinburgh in 2000.

Urine collection

A first morning urine sample of 25–50 mL was collected in sterile containers from all participants. Urine was kept at 4 °C until processing and centrifuged at 805×*g* for 5 min at 4 °C to remove cellular debris. Supernatant was aliquoted and frozen at −80 °C until analyzed. Urine creatinine was measured by laboratory standardized methods.

Untargeted metabolomic analysis: discovery stage

Untargeted metabolomic analysis of urine samples was performed using an Ultra-Performance Liquid Chromatography (UPLC) system coupled to an iFunnel Q-ToF (quadrupole-time-of-flight) Agilent 6550 mass spectrometer (Agilent Technologies, CA, USA) as previously described [22]. The metabolic differences between groups were identified

through a Volcano Plot analysis and significantly different variables (T-test P value < 0.05 and abs ($\log_2$ FC) > 1) were selected. Next, a supervised Pareto scaling OPLS-DA analysis with the variables was performed to evaluate whether they could differentiate between clinical groups. Finally, the most discriminant variables between groups were selected according to their VIP value (VIP > 1) and submitted to identification. Further details on materials and methods are available in the Online Supplementary Methods.

Targeted metabolomic analysis: validation stage

UPLC-MS/MS analysis

Urine sample preparation is detailed in the Online Supplementary Methods. The targeted analysis was performed using an Agilent 6460 high-performance liquid chromatography tandem mass spectrometer (triple quadrupole) (Agilent Technologies, CA, USA) followed by HESI in negative and positive modes. Metabolite separation was achieved using a UPLC BEH C18 column (100 × 2.1 mm, 1.7 µm) from Waters (Wexford, Ireland) with a gradient of mobile phases. Further details are described in the Online Supplementary Methods. Chromatographic conditions were optimized for an improved analytical response of spermine, adenosine, *p*-coumaric and *p*-cresol glucuronide.

Data and statistical analysis

The MassHunter Quantitative analysis (for QQQ) software (version B.07.00) was used for quantification. Final concentrations (µmol/mmol creatinine) were calculated considering the urine dilution factor and creatinine levels. A Kruskal–Wallis was used to evaluate differences in metabolites among the clinical groups studied. The U-Mann–Whitney test was used to evaluate differences in creatinine levels between BC and controls. In the validation stage, logistic regression models were adjusted using the expression level of the dysregulated metabolites. The predictive ability of the models was evaluated by estimating the area under the curve (AUC) for the receiver operator characteristics (ROC) analysis together with the best cut-off value to identify patients, sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) using MetaboAnalyst v.5.0. $P < 0.05$ was considered statistically significant.

Results

Clinical characteristics of the study subjects

BC patients and controls were divided in two cohorts for the discovery and the validation stages. For the discovery stage, we selected all TaG1 and TaT1G3 NMIBC and MIBC patients recruited at that moment. In the validation stage, we studied the additional patients recruited together with the subgroup of TaT1G2 NMIBC patients. The clinical characteristics of the study subjects are depicted in Table 1.

Continuous variables are presented as median and interquartile range (IQR). Categorical variables are presented as count and percentage. * Participants were clinically followed up until May 2020. BC, bladder cancer; MIBC, muscle-invasive bladder cancer; NMIBC, non-muscle-invasive bladder cancer; TURBT, transurethral resection of bladder tumor; BCG, Bacillus Calmette–Guerin; CIS, carcinoma in situ; NA, not available.

Untargeted metabolomic analysis: discovery of potential biomarkers

Metabolomic differences between all BC patients and healthy controls

First, we identified the metabolic differences between all BC patients (NMIBC + MIBC: TaG1 + TaG3 + T1G3 + T2G2 + T2G3 + T3G3) and healthy controls. Through the Volcano Plot analysis, 270 significantly different variables were selected (Figure S2a) and the supervised Pareto scaling OPLS-DA analysis revealed a clear separation between BC patients and controls (Fig. 1a) and the model showed good diagnostics and was satisfactorily validated (Fig. 1h). Table S1 depicts the 32 most discriminant variables. *p*-Cresol glucuronide was the only metabolite fully annotated (confidence level 1) with chemical standard showing and increased response (FC = 2.98, $P < 0.001$) in urine of BC patients (Table 2). We also found an increase in the 2 putatively annotated metabolites (confidence level 2), benzylamine and topiramate (FC = 2.13, $P < 0.001$; FC = 5.99, $P < 0.001$, respectively).

Metabolomic differences between TaG1 NMIBC patients and healthy controls

In a 2nd scenario, we compared the metabolomic profile of TaG1 NMIBC patients, those with the mildest grade, and healthy controls. The Volcano Plot showed 231 significantly different variables between both clinical groups (Figure S2b), which clearly separate the groups (Fig. 1b). The model was satisfactorily validated (Fig. 1h) and 21 variables were the most discriminant between groups (Table S2). From those, we were able to confirm with the chemical standard the decrease of *p*-coumaric acid (FC = 0.43, $P = 0.011$; confidence level 1) in urine of TaG1 BC patients (Table 2).

Table 1 Clinical characteristics of the BC patients and healthy controls studied

	BC		Controls	
	Discovery ($n=64$)	Validation ($n=134$)	Discovery ($n=20$)	Validation ($n=78$)
Age, years (IQR)	69 (64–77)	68 (62–74)	64 (52–66)	64 (56–68)
Male sex, *N* (%)	56 (87.5)	115 (85.8)	18 (90)	65 (83.3)
Urine creatinine, mg/dL (IQR)	90.6 (54.7–121.5)	74.0 (50.7–117.2)	91.2 (64.7–173.9)	105.2 (68.8–153.6)
Tumor stage and grade, *N* (%)				
TaG1	19 (29.7)	19 (14.2)	–	–
TaG2	–	62 (46.3)	–	–
TaG3	11 (17.2)	9 (6.7)	–	–
T1G2	–	15 (11.2)	–	–
T1G3	20 (31.2)	16 (11.9)	–	–
T2G2	1 (1.6)	4 (3)	–	–
T2G3	12 (18.7)	9 (6.7)	–	–
T3G3	1 (1.6)		–	–
Type of surgery, *N* (%)				
TURBT	53 (82.8)	118 (88.1)	–	–
Cystectomy	11 (17.2)	16 (11.9)	–	–
Disease status until the end of the follow-up*, *N* (%)				
Disease-free alive	42 (65.6)	103 (76.8)	–	–
Progression alive	5 (7.8)	2 (1.5)	–	–
Death by BC	9 (14.1)	16 (12)	–	–
Death by other	8 (12.5)	13 (9.7)	–	–
Tumor origin, *N* (%)				
Primary	38 (59.4)	77 (57.5)	–	–
Recurrent	26 (40.6)	57 (42.5)	–	–
MIBC progression from NMIBC, *N* (%)	3 (4.7)	4 (3.0)	–	–
Hematuria, *N* (%)				
At diagnosis	46 (34.3)	70 (52.2)	–	–
At sampling	33 (51.6)	57 (42.5)	0 (0)	0 (0)
Tumor size, cm^2				
<1	5 (7.8)	16 (11.9)	–	–
1–3	38 (59.4)	80 (59.7)	–	–
>3	21 (32.8)	38 (28.4)	–	–
Multifocality, *N* (%)				
0	34 (53.1)	8 (6.0)	–	–
1	9 (14.1)	73 (54.4)	–	–
2–3	15 (23.4)	32 (23.9)	–	–
>3	6 (9.4)	21 (15.7)	–	–
Previous treatment, *N* (%)				
Doxorubicin	0 (0.0)	3 (2.2)	–	–
Gemcitabine	0 (0.0)	1 (0.7)	–	–
Mitomycin	1 (1.6)	8 (6.0)	–	–
BCG	9 (14.1)	7 (5.2)	–	–
Tumor histology				
Papillary	59 (92.2)	126 (94.0)	–	–
Concomitant CIS	5 (7.8)	8 (6.0)	–	–
Smoking, *N* (%)				
No	12 (18.8)	29 (21.6)	NA	NA
Former	24 (37.5)	53 (39.6)	NA	NA
Current	25 (39.0)	46 (34.3)	NA	NA

Table 1 (continued)

	BC		Controls	
	Discovery ($n=64$)	Validation ($n=134$)	Discovery ($n=20$)	Validation ($n=78$)
Unknown	3 (4.7)	6 (4.5)	NA	NA

Metabolomic differences between TaG3 + T1G3 NMIBC patients and healthy controls

In a 3rd scenario, we compared the metabolomic profile of TaG3 + T1G3 NMIBC patients, those with NMIBC patients with the worse prognosis, and healthy controls. The Volcano Plot showed 315 significantly different variables between groups (Figure S2c) that clearly separate the groups (Fig. 1c). The model was satisfactorily validated (Fig. 1h) and 16 variables were the most discriminant between groups (Table S3). We found that TaG3 + T1G3 NMIBC patients excrete in urine higher levels of 2 metabolites with confidence level 2 (Table 2).

Metabolomic differences between MIBC patients and healthy controls

In a 4th scenario, we compared the metabolomic profile of all MIBC patients (T2G2 + T2G3 + T3G3), the most aggressive BC, and healthy controls. The Volcano Plot showed 341 significantly different variables between both groups (Figure S2d) that clearly separate the groups (Fig. 1d). The model was satisfactorily validated (Fig. 1h) and 33 variables were the most discriminant between groups (Table S4). By comparison with a chemical standard, we could confirm that MIBC patients excrete in urine lower levels of spermine (FC = 0.34, $P = 0.028$) and adenosine (FC = 0.49, $P < 0.001$) (Table 2).

Metabolomic differences between NMIBC patients and healthy controls

In a 5th scenario, we compared the metabolomic profile of all NMIBC patients (TaG1 + TaG3 + T1G3) and healthy controls. The Volcano Plot showed 272 significantly different variables between both groups (Figure S2e) that clearly separate the groups (Fig. 1e). The model was satisfactorily validated (Fig. 1h). From the 24 variables selected as the most discriminant between groups (Table S5), only *p*-cresol glucuronide was fully confirmed with the chemical standard and a decrease observed in urine of healthy control (FC = 3.21, $P = 0.001$) (Table 2).

Metabolomic differences between TaG1 and TaG3 + T1G3 NMIBC patients

In a 6th scenario, we compared the metabolomic profile of the NMIBC patients with the mildest grade (TaG1) and that of the NMIBC patients with the worse prognosis (TaG3 + T1G3). The Volcano Plot showed 102 significantly different variables (Figure S2f) that clearly separate both groups (Fig. 1f). The model was satisfactorily validated (Fig. 1f) and 10 variables were selected as the most discriminant between groups (Table S6).

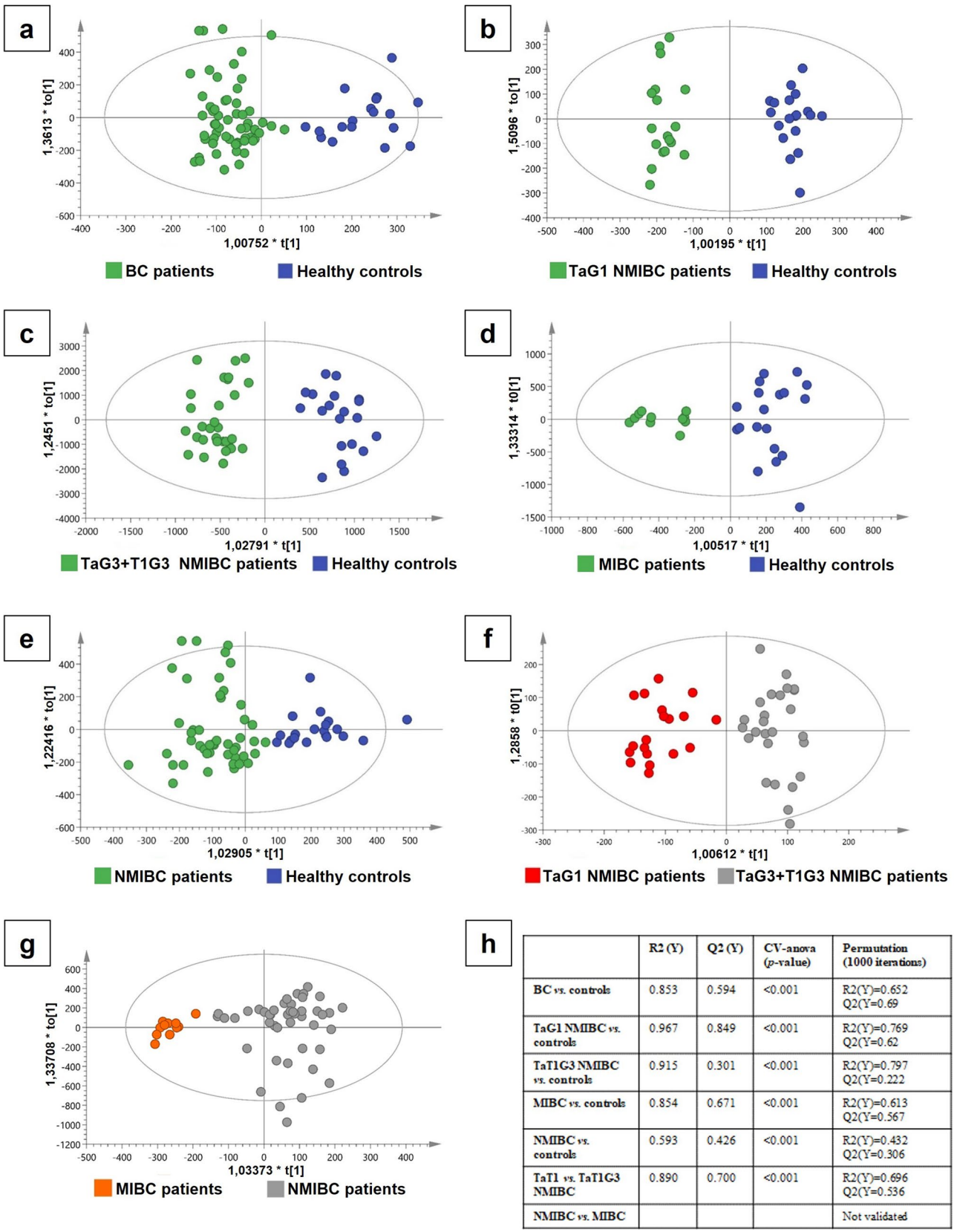
Metabolomic differences between NMIBC and MIBC patients

Lastly, in a 7th scenario, we compared the metabolomic profile of all NMIBC and all MIBC patients. The Volcano Plot showed 215 significantly different variables between the two groups (Figure S2g); however, the model could not be validated (Fig. 1h). Taking a closer look at the results (Fig. 1g), the MIBC patients form a tight cluster but the NMIBC patients show a high dispersion. This indicates that, far from being a homogenous entity, NMIBC comprises tumors with different levels of complexity that hampers the identification of metabolic variables able to discriminate between NMIBC and MIBC patients. As a consequence, no further analysis was performed in this scenario.

Targeted metabolomic analysis: validation of potential biomarkers

Next, in an independent cohort of 134 BC patients and 78 controls we validated the dysregulation of metabolites with a LC–MS/MS targeted method developed. In this stage, we included a new subgroup of intermediate TaT1G2 NMIBC patients.

In the validation cohort, we evidenced a great increase of *p*-cresol glucuronide in urine of all categories of BC patients (median 3.5 µmol/mmol creatinine, IQR 0.3–1.5) compared to controls (0.7 µmol/mmol creatinine, 0.3–1.5) (Fig. 2a). At this validation stage, we obtained a ROC curve AUC of 0.79 [95% confidence interval (CI) (0.73–0.842)] (Fig. 2b). The best cut-off value of *p*-cresol glucuronide to identify BC patients from controls was 1.44 µmol/mmol creatinine, achieving a 70% sensitivity and a 70% specificity, and a PPV



h

	R2 (Y)	Q2 (Y)	CV-anova (*p*-value)	Permutation (1000 iterations)
BC *vs.* controls	0.853	0.594	<0.001	R2(Y)=0.652 Q2(Y=0.69
TaG1 NMIBC *vs.* controls	0.967	0.849	<0.001	R2(Y)=0.769 Q2(Y=0.62
TaT1G3 NMIBC *vs.* controls	0.915	0.301	<0.001	R2(Y)=0.797 Q2(Y=0.222
MIBC *vs.* controls	0.854	0.671	<0.001	R2(Y)=0.613 Q2(Y=0.567
NMIBC *vs.* controls	0.593	0.426	<0.001	R2(Y)=0.432 Q2(Y=0.306
TaT1 *vs.* TaT1G3 NMIBC	0.890	0.700	<0.001	R2(Y)=0.696 Q2(Y=0.536
NMIBC *vs.* MIBC				Not validated

◂**Fig. 1** Metabolomic differences between BC patients and healthy controls. OPLS-DA score plot of these different variables in **a** All BC patients and healthy controls; **b** TaG1 NMIBC patients and healthy controls; **c** NMIBC (TaG3 + T1G3) patients and healthy controls; **d** MIBC patients and healthy controls; **e** NMIBC patients and healthy controls; **f** TaG1 and TaG3 + T1G3 NMIBC patients; **g** NMIBC and MIBC patients; **h** Validation of the models with a sevenfold cross validation method (CV ANOVA) and permutation tests

of 83% and a NPV of 63%. The highest levels of *p*-cresol glucuronide were present in TaG1 NMIBC (6.7 µmol/mmol creatinine, 2.9–18.4), and the ROC curve AUC when compared NMIBC to controls was 0.803 (0.738–0.857) (Fig. 2c). The best cut-off value of *p*-cresol glucuronide to identify NMIBC patients from controls was 1.79 µmol/mmol creatinine, achieving an 80% sensitivity and a 70% specificity, and PPV of 85% and a NPV of 62%. However, no differences arose among BC subcategories, what suggests that *p*-cresol glucuronide may be a good non-invasive diagnostic marker for BC.

To validate the reduction of cis-*p*-coumaric acid in TaG1 NMIBC compared to controls, the trans-*p*-coumaric acid standard had to be used as the cis version was commercially unavailable. Consequently, the peak obtained in urine samples of the validation cohort did not coincide with the trans version and no further analysis could be performed to validate this metabolite.

In the validation cohort, spermine was increased in all BC subtypes (0.27 µmol/mmol creatinine, 0.16–0.56) compared to controls (0.15 µmol/mmol creatinine, 0.10–0.29) (Fig. 3a) and the greatest increase occurred in MIBC patients (0.58 µmol/mmol creatinine, 0.36–1.48). At this validation stage, we obtained a ROC curve AUC of 0.882 (0.774–0.958) (Fig. 3b). The best cut-off value of spermine to identify MIBC patients from controls was 0.33 µmol/mmol creatinine, achieving an 80% sensitivity and an 80% specificity, and PPV of 44% and a NPV of 97%. When spermine levels were compared among BC subgroups, significant differences only arose between MIBC and TaT1G2 NMIBC (0.20 µmol/mmol creatinine, 0.13–0.48) ($P = 0.0214$), probably because this subgroup of NMIBC contains the greatest number of patients.

Finally, the decrease in adenosine levels observed in MIBC patients compared to controls in the discovery cohort, could not be validated (Fig. 4) as similar adenosine levels were found in all categories studied.

Thirteen of the 198 patients studied (6.6%) had concomitant carcinoma in situ (CIS): 2 TaG3 (15.4%), 7 T1G3 (53.8%) 2 T2G3 (15.4%) and 2 TaG2 (15.4%). No differences in *p*-cresol glucuronide (2.733 *vs.* 3.542 µmol/mmol creatinine, $P = 0.2774$), spermine (0.4919 *vs.* 0.2609 µmol/mmol creatinine, $P = 0.1807$) or adenosine (0.02360 *vs.* 001,710 µmol/mmol creatinine, $P = 0.5690$) levels were observed between patients with concomitant CIS and those without.

Discussion

Tumor development and proliferation is dependent, in part, on metabolic reprogramming to support the increased energy and biosynthetic demands of the malignant phenotype. Accordingly, these metabolic changes may result in unique urine metabolite signatures [23, 24], as in BC patients.

Herein, we have analyzed for the first time the urine metabolomic profile of NMIBC and MIBC patients of different stages and that of healthy controls to identify novel urine metabolites with staging and diagnostic purposes. Furthermore, we aimed to verify the identity of these metabolites using chemical standards, focusing on metabolites that could have a clinical implication in BC development or prognosis, thus discarding drug metabolites, dietary components, or food additives. In a discovery stage, we performed a series of comparisons between the different clinical groups, and we found dysregulated metabolites in all comparisons but NMIBC *vs.* MIBC (Tables S1- S6). While MIBC patients were remarkably homogeneous, NMIBC patients were very dissimilar, which is confirmed by the identification of 10 dysregulated metabolites between TaG1 and TaT1G3 NMIBC patients. Our results corroborate that NMIBC is a heterogeneous entity that comprises tumors with a wide range of complexity. We also attempted to verify the identity of as many metabolites as possible that we carefully selected based on their probable clinical relation to BC. However, standards for 6-hydroxy-5-methoxyindole glucuronide, norepinephrine sulfate, 3-hydroxydecanoyl carnitine, cholestane-3,7,12,25-tetrol-3-glucuronide, 6-ketodecanoylcarnitine, 9-decenoylcarnitine and dihydroferulic acid 4-O-glucuronide were commercially unavailable at the moment. Lastly, in an independent cohort of BC patients and controls, using a targeted approach, we aimed to validate the differential expression of *p*-cresol glucuronide, *p*-coumaric acid, spermine and adenosine in urine of BC patients. The levels of these three metabolites in urine were not affected by the concomitant presence of CIS.

We validated the increase of *p*-cresol glucuronide in urine of all BC patients and more pronouncedly in that of NMIBC patients. *p*-Cresol glucuronide is an endogenous product that is characteristically excreted in urine; however, at high concentrations cannot be cleared by the kidneys and accumulate becoming uremic toxins. Many of these highly concentrated toxins are gut-derived metabolites generated via protein fermentation by bacteria in the colon [25], particularly dietary tyrosine is converted by colonic microbiota to *p*-cresol that is absorbed into hepatic circulation, where it is metabolized by UDP-glucuronosyltransferase into *p*-cresol glucuronide

Table 2 Identification of the most discriminant metabolomic variables with confidence level 1 or 2 obtained in the different comparisons performed

Comparison	AM (*m/z*)	Rt (min)	Identification of variables		*P* value	Trend
			Metabolite annotation	Confidence level		
BC *vs.* HC	307.079	4.580	*p*-Cresol glucuronide	1	< 0.001	↑
BC *vs.* HC	108.081	0.827	Benzylamine	2	< 0.001	↑
BC *vs.* HC	340.107	2.499	Topiramate	2	< 0.001	↑
TaG1 NMIBC *vs.* HC	147.044	3.3123	*cis-p*-Coumaric acid	1	0.011	↓
TaG1 NMIBC *vs.* HC	108.081	0.827	Benzylamine	2	0.002	↑
TaG1 NMIBC *vs.* HC	340.107	2.499	Topiramate	2	0.005	↑
TaG1 NMIBC *vs.* HC	377.088	4.76	2′,3,5-Trihydroxy-5′,7-dimethoxyflavanone*	2	0.005	↓
TaG3 + T1G3 NMIBC *vs.* HC	507.149	5.554	7-Methyl-1,4,5-naphthalenetriol 4-[xylosyl-(1- > 6)-glucoside]	2	0.016	↑
TaG3 + T1G3 NMIBC *vs.* HC	597.364	6.893	L-Urobilinogen	2	0.018	↑
MIBC *vs.* HC	203.224	0.6	Spermine	1	0.028	↓
MIBC *vs.* HC	268.103	1.610	Adenosine	1	< 0.001	↓
MIBC *vs.* HC	135.044	4.3685	R-(-)-Mandelic acid	2	0.032	↓
MIBC *vs.* HC	300.218	5.9423	3-Hydroxynonanoyl carnitine	2	0.003	↓
MIBC *vs.* HC	310.202	5.6923	Metipranolol Nadolol	2	0.006	↓
MIBC *vs.* HC	343.223	5.2919	18-Hydroxy-5Z,8Z,11Z,14Z-eicosatetraenoic acid*	2	< 0.001	↓
MIBC *vs.* HC	411.163	5.7576	7-Epi-12-hydroxyjasmonic acid glucoside*	2	0.016	↓
MIBC *vs.* HC	457.205	5.92	Delavirdine	2	0.003	↓
MIBC *vs.* HC	529.264	5.9890	(3b,9R)-5-Megastigmene-3,9-diol 9-[apiosyl-(1- > 6)-glucoside]*	2	0.015	↓
NMIBC *vs.* HC	307.079	4.580	*p*-Cresol glucuronide	1	0.001	↑
NMIBC *vs.* HC	108.081	0.827	Benzylamine	2	< 0.001	↑
NMIBC *vs.* HC	340.107	2.499	Topiramate	2	0.001	↑

HC healthy controls, *AM* accurate mass, *RT* retention time, Confidence level: 1: confirmed; 2: putatively annotated

*Additional annotated metabolites are detailed in Table S1

and by sulfotransferase into *p*-cresol sulfate [26]. *p*-Cresol metabolites like *p*-cresyl glucuronide accumulate in patients with chronic kidney disease and might contribute to disease progression [27]. In fact, we recently described *p*-cresol glucuronide as a potential urine biomarker for the diagnosis of renal cell carcinoma [22]. In line with our results, Pasikanti et al. [28] observed an increase of *p*-cresol glucuronide in urine of BC patients compared to non-bladder cancer controls, but no additional studies have identified this metabolite in urine of BC patients. While the increase in *p*-cresol glucuronide is comprehensible in urine of patients with chronic kidney disease and renal cell carcinoma, the origin of the increase in this metabolite in urine of BC patients remains unsolved. Cresols are tumor promoters in rodents [29] and the oral administration of saccharin, which increases protein and amino acid substrates, resulted in perturbed metabolism of amino acids by gut flora, leading to an increased formation of co-carcinogens like cresol [30]. Additional studies should be conducted to ascertain the involvement of *p*-cresol glucuronide in human BC carcinogenesis. The decrease of *p*-coumaric acid found in urine of the mildest form of NMIBC could not be validated due to the unavailability of the chemical standard. Future studies should be conducted to validate its usefulness as staging biomarker for TaG1 NMIBC once the cis version of the standard is commercially available.

Finally, we validated the increase of spermine in urine of all BC patients and more pronouncedly in that of MIBC patients. High levels of urinary polyamines have been long related to solid tumors [12, 31, 32] and have been shown to correlate with poor prognosis [33]. The increase in urine polyamines in BC was described long ago [34]. Polyamines are ubiquitous molecules with multiple intracellular functions. Almost all cells can produce polyamines (including putrescine, spermidine and spermine) and their production is especially high in rapidly growing cells, like cancer cells, because polyamines are indispensable for cell growth. In fact, bladder epithelial cells isolated from rats require the addition of putrescine, spermine, and spermidine to attain maximum growth and long-term survival

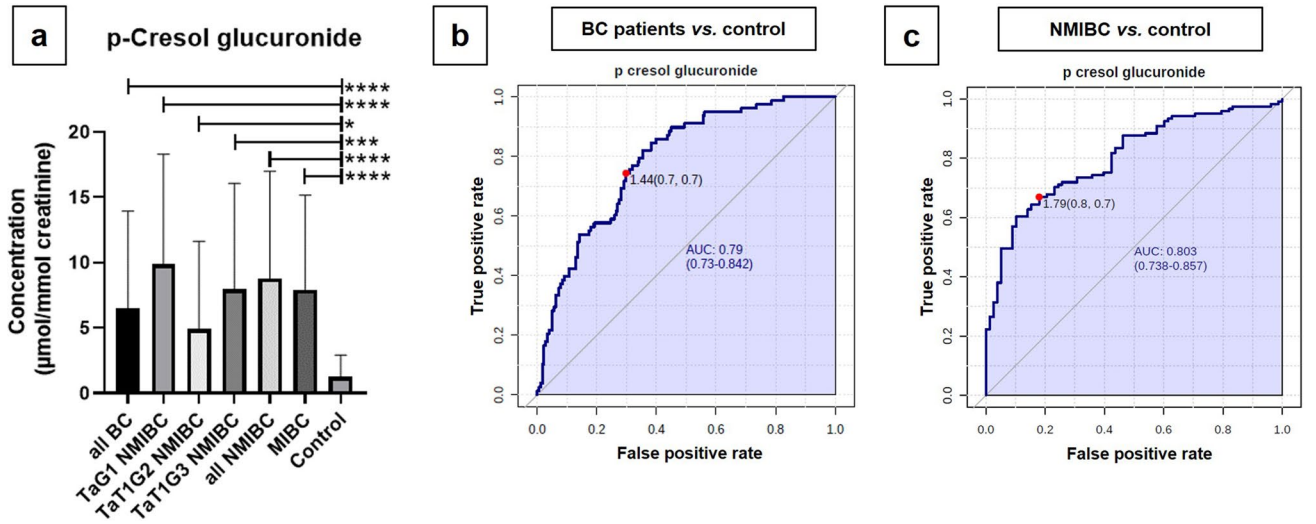


Fig. 2 Validation of *p*-cresol glucuronide in BC patients and controls. **A** Urine concentration of *p*-cresol glucuronide; **B** Validated ROC curve of *p*-cresol glucuronide comparing all BC patients *vs.* controls; **C** Validated ROC curve of *p*-cresol glucuronide comparing NMIBC patients *vs.* controls. $^{*}P<0.05$; $^{***}P<0.001$; $^{****}P<0.0001$

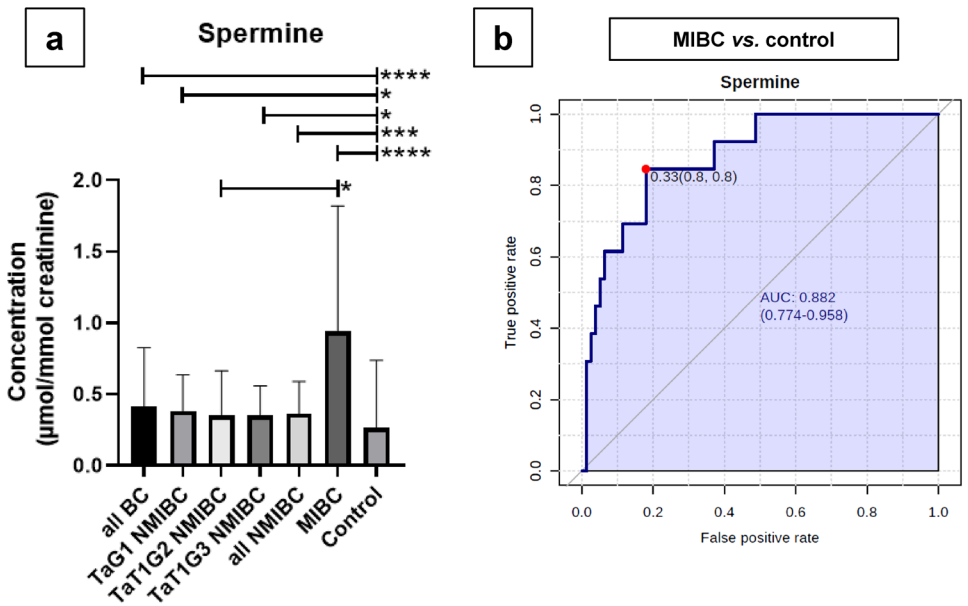


Fig. 3 Validation of spermine in BC patients and controls. **A** Urine concentration of spermine; **B** Validated ROC curve of spermine comparing MIBC patients *vs.* controls. $^{*}P<0.05$; $^{***}P<0.001$; $^{****}P<0.0001$

[35]. Cells tightly regulate their levels through feedback mechanisms affecting synthesis, intracellular conversion, and transport. Because polyamines have an important role in regulating cell growth, they are a target for cancer therapeutic development and cancer prevention [36]. In fact, the spermine analog N1,N11-diethylnorspermine (DENSPM) has been shown to induce the polyamine-acetylating enzyme spermidine/spermine N^1-acetyltransferase (SAT1), disrupt polyamine pool homeostasis, and inhibit BC tumor growth [37]. However, to effectively inhibit cell growth through polyamine depletion, one needs to inhibit both polyamine synthesis and import [38] as inhibition of endogenous polyamine biosynthesis results in compensatory up-regulation of polyamine uptake. Increased polyamine availability enhances the capability of cancer cells to invade and metastasize to new tissues while diminishing immune cells' anti-tumor immune functions [39]. As a high rate of cell growth and death occurs in tumor tissues, the highest levels of spermine observed in urine of our MIBC patients may be derived from bladder tumor cells where an increased polyamine biosynthesis and probably a concomitant increase of polyamine uptake may be occurring to promote cell growth. This would reinforce

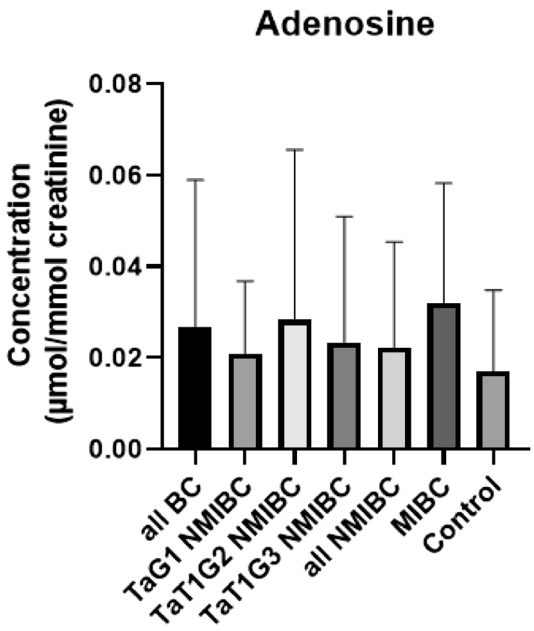


Fig. 4 Urine concentration of adenosine in BC patients and controls

the functional role of spermine in BC progression and invasiveness of this aggressive subtype.

Given that BC diagnosis is usually belated since symptoms are often unspecific and that cytology is unspecific for low-grade tumors, urine metabolomic profiling may clearly improve patient counseling, particularly identifying distinct tumor stage/grade groups, while reducing patient discomfort. Particularly, urine *p*-cresol glucuronide profiling may improve early BC diagnosis and NMIBC staging, which patients may benefit the most as this would clearly result in a better patient performance and tumor remission. In addition, urine spermine profiling may promote early MIBC staging, and this would enable to start an earlier therapeutic approach to promote a better patient performance and, maybe, reduce the odds of tumor relapse.

A limitation of our study may be the sample size of certain BC stages studied; thus, the different stage/grade groups are misbalanced. Future studies in larger balanced cohorts will improve the predictive values of the identified metabolites, especially the PPV of spermine. Transferability into daily clinical practice may not be a limitation as tertiary-level hospitals have the means to conduct these measurements, currently employed to diagnose metabolopathies following similar protocols, and it would be clearly faster, cheaper and safer than cystoscopies. Strengths of our study are a thorough clinical evaluation of patients, the study of all stages of BC patients (especially MIBC patients), the inclusion of healthy volunteers with clinical evidence of absence of urological disorders, the verification of dysregulated metabolites with chemical standards and the validation of dysregulated metabolites in an independent cohort of BC patients. Future steps comprise the confirmation of our findings in external independent BC cohorts, preferably recruited in different countries, and the set-up of our experimental approach in the clinical circuit of hospital laboratories. In addition, the validation of *p*-coumaric acid as TaG1 NMIBC staging biomarker should be performed once the cis version of the standard is commercially available. These studies will reinforce our findings and will evaluate the status of these metabolites when tumor relapse occurs. In addition, these studies will reveal the performance of markers like *p*-cresol glucuronide and spermine in BC diagnosis and staging since these are also dysregulated in other tumor types.

Conclusion

Our work is the first metabolomics study conducted in urine of a thoroughly characterized cohort comprising the different stages of NMIBC, MIBC and healthy controls. We have identified urine *p*-cresol glucuronide as potential diagnostic biomarker for BC and as staging biomarker for NMIBC patients. Among NMIBCs, *p*-coumaric acid may be a potential specific staging biomarker for the mildest form of NMIBC (TaG1); however, future validation experiments should be conducted once the precise version of the standard is commercially available. Remarkably, for MIBC, we have identified spermine as potential specific staging biomarker.

Our findings highlight the strong potential of urine metabolomic profiles as non-invasive biomarkers for BC. These may improve current weaknesses in daily clinical practice, such as the use of harmful and highly invasive techniques to diagnose and frequent monitor BC, the low sensitivity of these techniques in low-grade tumors and the enormous medical expenses entailed.

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Author contributions JO: protocol development, data collection, data analysis and manuscript writing. ÁFP: data analysis. MR: protocol development, data analysis, and manuscript writing. EP: manuscript editing and supervision. FC: protocol development. RH: data analysis. JPA: resources. CDVD: resources and manuscript editing. MMS: resources and manuscript editing. PM: project development, management, data analysis, manuscript writing and funding acquisition. All authors read and approved the final manuscript.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval and consent to participate Informed consent was obtained from all participants according to protocols approved by the ethics review board at La Fe University and Polytechnic Hospital. The study was performed according to the declaration of Helsinki, as amended in Edinburgh in 2000.

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